9.1.6 Adding more elements in a list

In Python, append( ) function is used to add a single element and extend( ) function is used to add more than one element to an existing list.

Syntax:

List.append (element to be added)
List.extend ([elements to be added])

In extend( ) function, multiple elements should be specified within square bracket as arguments of the function.

Example

```
>>> Mylist=[34, 45, 48]
>>> Mylist.append(90)
>>> print(Mylist)
        [34, 45, 48, 90]
```

In the above example, Mylist is created with three elements. Through >>> **Mylist.append(90)** statement, an additional value 90 is included with the existing list as last element, following print statement shows all the elements within the list MyList.

Example

```
>>> Mylist.extend([71, 32, 29])
>>> print(Mylist)
        [34, 45, 48, 90, 71, 32, 29]
```

In the above code, extend( ) function is used to include multiple elements, the print statement shows all the elements of the list after the inclusion of additional elements.

9.1.7 Inserting elements in a list

As you learnt already, append( ) function in Python is used to add more elements in a list. But, it includes elements at the end of a list. If you want to include an element at your desired position, you can use insert ( ) function. The insert( ) function is used to insert an element at any position of a list.

Syntax:

List.insert (position index, element)

Example

```
>>> MyList=[34,98,47,'Kannan', 'Gowrisankar', 'Lenin', 'Sreenivasan' ]
>>> print(MyList)
        [34, 98, 47, 'Kannan', 'Gowrisankar', 'Lenin', 'Sreenivasan']
>>> MyList.insert(3, 'Ramakrishnan')
>>> print(MyList)
[34, 98, 47, 'Ramakrishnan', 'Kannan', 'Gowrisankar', 'Lenin', 'Sreenivasan']
```

In the above example, insert( ) function inserts a new element 'Ramakrishnan' at the index value 3, ie. at the 4th position. While inserting a new element in between the existing elements, at a particular location, the existing elements shifts one position to the right.